



COMP 234 - Databases

Midterm

February 20th – February 27th (Take-Home)

Instructor: Zach Horvath

Pages: 5 (Including cover page)

This is an open book exam. You may use any resources you require **other** than AI, chat bots, chat rooms, messaging software, discussion forums or other communication technology. If you are caught using communication technology listed above or otherwise, you will be asked to leave the exam and receive a 0 on this assessment.

Academic dishonesty will NOT be tolerated. This exam is an individual assessment. **It should be your own work.** Sharing the contents of this exam or trying to use the answers provided by others who have shared this exam will not be tolerated. Sharing your own work or using the work of others will not be tolerated. If you do not submit your own work, you will receive a 0% on this assessment. If it is suspected you did not submit your own work, the department reserves the right to conduct an interview to assess your knowledge of your submission.

At the end of the exam, submit this exam to your instructor using BrightSpace; failure to submit your exam will result in a 0% on your exam.

Grade Breakdown		
Section	Marks	
Data Definition and Manipulation Language		12
Entity Relationship Diagrams		8
Data Query Language		25
Total:		45



General steps for your exam:

- Create a new Local Database called Midterm2025.
- Using the new Midterm Database, run the SQL Script (on Brightspace) called CreateDB.sql
- Provide all answers in a .sql file. Separate your questions with a comment, ie:
-- Question 1: here is my answer
- There is an ERD that may be helpful included in this exam.

Data Definition and Manipulation Language

- 1) Write the SQL to create a new table called *Logins*. This table will hold the login formation for our employees & it should follow the following entity [3 marks]:

Logins		
PK	ID	INT
	UserName	varchar(100) NOT NULL
	PasswordHash	Binary(128) NULL
	LastLogin	Datetime NULL
FK	EmployeeID	INT

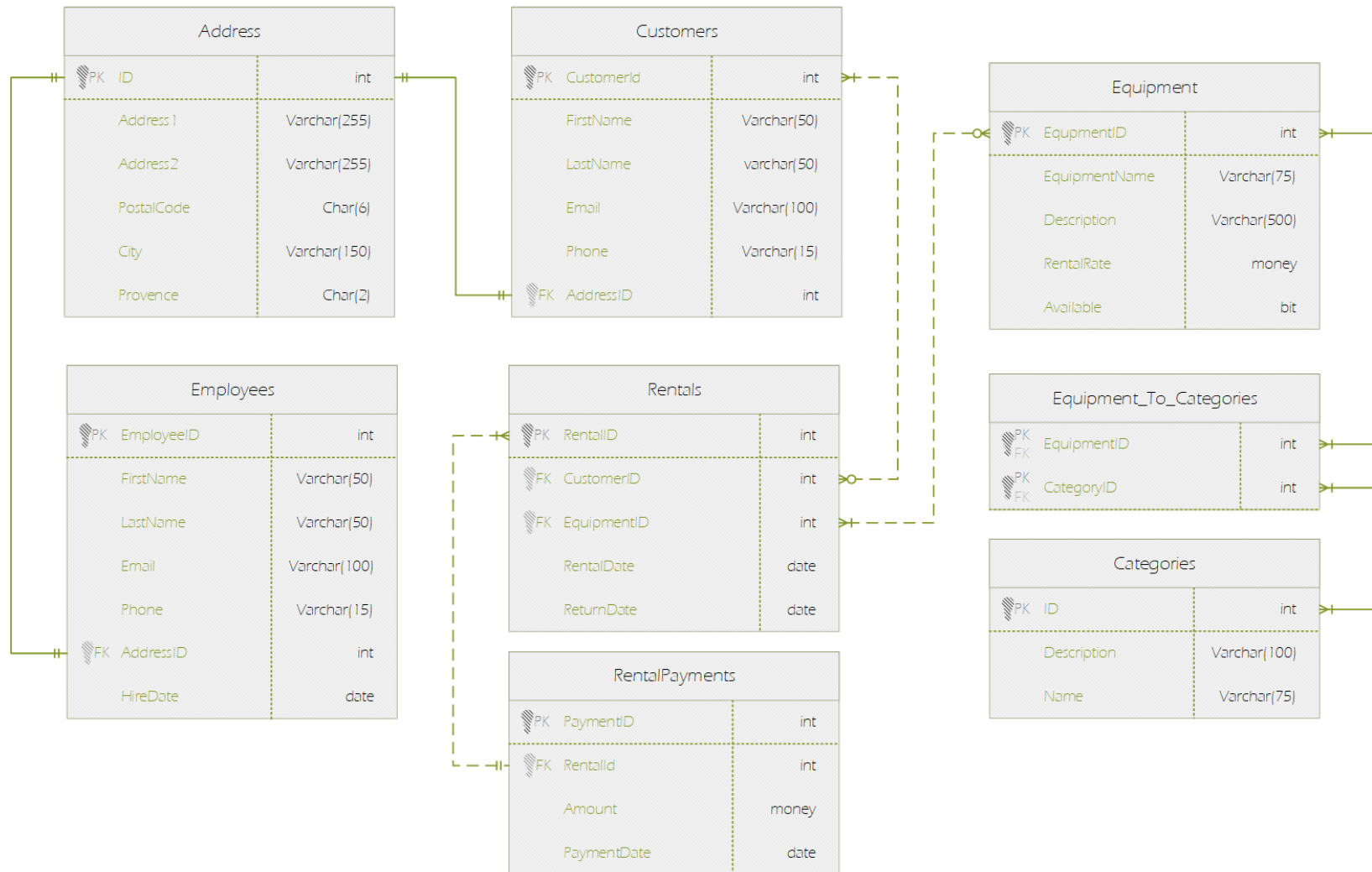
Note*: EmployeeID is a Foreign Key to EmployeeID in the Employees table.

2) The company's first employee, John Smith, is finally retiring. Write the SQL command to remove them from the Employee table and be sure to identify them using the primary key.

[2 marks]

- 3) Looks like there is now a job available! Add yourself to the Employee table **and any required parent tables**. Use your name, Saskpolytech email, your student ID as your phone number (no need to add any phone formatting), and the current date (provided by a function) as HireDate. Answer in a comment: what other table did you need to add a record to and why? [5 marks]
- 4) Our Customer Christin Gonzalez who lives on 999 Maple Rd was just married! Write the SQL command to update her last name to 'Ferguson' [2 mark]

Entity Relationship Diagrams



Entity Relationship Diagrams – Continued

Please indicate your answers to the following questions:

- 5) Add the Login table you created to the ERD, 1 employee has 1 and only 1 login [3 marks].
- 6) What is the cardinality between Equipment and Rentals? _____ to _____, [2 mark]
- 7) What is the cardinality between Customers and Address? _____ to _____. [2 mark]
- 8) The keys in Equipment_To_Categories are an example of a (circle) [1 mark]:
- a. Comparing Keys
 - b. Composite Keys
 - c. Companion Keys
 - d. None of the Above

Data Query Language

For these queries, be sure to provide a column alias where appropriate and if a sample is given with formatting, match that sample for full marks.

- 9) We're looking to highlight our collection of Saws in our next flyer. Write a query that will produce a result set containing every piece of equipment with saw in the name. Don't include saws with an availability of 0, and order by rental rate descending. Your result set should look like the following (notice column names have spaces), but will have more results [5 marks]:

Equipment Name	Rental Rate	Available
Masonry Saw	85.00	1
Tile Saw	69.99	1
...	...	...

10) We're curious how business was in 2023. Provide the total amount of RentalPayments we received in 2023. Your results set should look like the following [3 marks]:

2023 Revenue
\$XX,XXX.XX

- Let's break this down more. Write a query to provide the date and the amount of the largest payment recorded. (Hint – use a nested subquery) [3 marks]
- Write a query to find how many RentalPayments ever made were greater than the average historical payment? [3 marks]

11) We're going to produce a mailer for our customers! Write a query that returns our customer FirstName, LastName combined as Full Name. **Use an inner join** to return their address1, address2, city, postalCode and province **only** for customers from AB. Your results should look similar to [5 marks]:

Full Name	Address1	Address2	City	Postal Code	Province
Michael Brown	555 Pine St	Unit B	Calgary	T4T4T4	AB
Emily Davis	777 Cedar St	NULL	Calgary	R6R6R6	AB
Jessica Anderson	222 Spruce Ave	NULL	Calgary	S9S9S9	AB
Ashley Garcia	333 Oak Rd	NULL	Calgary	P1P1P1	AB

12) We need a way to provide estimates to our customers. Currently Rental Rates is calculated by the application that runs our database. Write a query to get the difference in days between RentalDate and ReturnDate (You'll need a function for this) and then using an inner join multiply the day(s) by the RentalRate for that piece of equipment. For example [6 marks]:

RentalID 1 is has 5 days between RentalDate and ReturnDate

EquipmentID 5's RentalRate is \$62.80

If you're WHERE clause had RentalID =1 your result set would look like this:

Cost
\$314.00

If we use RentalId = 13 it would return:

Cost
\$227.5